

DATA

DATA TYPES

Fabio Stella

Associate Professor

c/o Department of Informatics, Systems and Communication

University of Milano Bicocca



DATA TYPES

The following concepts will be introduced:

- ✓ **DATA SET**
- ✓ **ATTRIBUTE**
- ✓ **TYPE OF ATTRIBUTES**
 - NOMINAL
 - ORDINAL
 - INTERVAL
 - RATIO
 - DISCRETE
 - CONTINUOUS

Assume you work in a **DATA MINING COMPANY** while a friend of yours is the **CHIEF EXECUTIVE OFFICER** of a **TELECOMMUNICATION COMPANY**.



Your friend read an article where it is stated that **DATA MINING HELPS TO MAKE INFORMED AND ACTIONABLE DECISIONS** in the Retail Sector.



Your friend is curious to know from you **WHETHER USING THE DATA MINING METHODOLOGY IT IS POSSIBLE TO EXTRACT KNOWLEDGE FROM DATA** to help **MAKING EFFECTIVE DECISIONS IN THE TELECOM SECTOR**.



You tell your friend that you are available to analyze the telecom data to prove that **DATA MINING IS EFFECTIVE TO EXTRACT VALUABLE/ACTIONABLE KNOWLEDGE FROM DATA**.

Your friend thanks you and promises to **SEND YOU**, as soon as will be back to the office, **A FILE CONTAINING THE DATA WHICH ARE RELEVANT TO SOLVE THE CHURN PROBLEM, THE PROBLEM OF DISCOVERING WHICH ARE THE UNFAITHFUL CUSTOMERS (CHURNERS)**.

After two days you receive an **EMAIL MESSAGE FROM YOUR FRIEND** with attached a txt file named churn.



You download and open the **CHURN.TXT FILE AND INSPECT THE FIRST 4 COLUMNS AND THE FIRST 10 LINES:**

Area Code	Day Mins	Eve Mins	Churn
415	?	197.4	n
415	161.6	195.5	n
415	?	121.2	n
408	299.4	61.9	n
415	166.7	148.3	y
510	223.4	220.6	n
510	218.2	348.5	n
415	157	103.1	n
408	184.5	351.6	n
415	258.6	222	n

Area Code

code of the customer's area

Day Mins

minutes of the day calls

Eve Mins

minutes of the evening calls

Churn

does the customer churned?

{n, y}

After two days you receive an **EMAIL MESSAGE FROM YOUR FRIEND** with attached a txt file named churn.



You download and open the **CHURN.TXT FILE AND INSPECT THE FIRST 4 COLUMNS AND THE FIRST 10 LINES:**

Area Code	Day Mins	Eve Mins	Churn
415	?	197.4	n
415	161.6	195.5	n
415	?	121.2	n
408	299.4	61.9	n
415	166.7	148.3	y
510	223.4	220.6	n
510	218.2	348.5	n
415	157	103.1	n
408	184.5	351.6	n
415	258.6	222	n

You notice that the value of column **DAY MINS** for the **FIRST AND THIRD RECORDS** takes the **SUSPECT VALUE** of **"?"**.

You ask your friend about **THE "?" VALUE**, who replies that it is the value which is used to mean that the value of the field is **MISSING**, i.e., it has not been recorded.

After two days you receive an **EMAIL MESSAGE FROM YOUR FRIEND** with attached a txt file named churn.



You download and open the **CHURN.TXT FILE AND INSPECT THE FIRST 4 COLUMNS AND THE FIRST 10 LINES:**

Area	Code	Day	Mins	Eve	Mins	Churn
415			?	197.4		n
415			161.6	195.5		n
415			?	121.2		n
408			299.4	61.9		n
415			166.7	148.3		y
510			223.4	220.6		n
510			218.2	348.5		n
415			157	103.1		n
408			184.5	351.6		n
415			258.6	222		n

File **DATA SET**

Column **ATTRIBUTE**; property or characteristic of an object that may vary, either from one object to another or from one time to another.

Row **RECORD** or **CASE** or **OBSERVATION**

After two days you receive an **EMAIL MESSAGE FROM YOUR FRIEND** with attached a txt file named churn.



You download and open the **CHURN.TXT FILE AND INSPECT THE FIRST 4 COLUMNS AND THE FIRST 10 LINES:**

Area Code	Day Mins	Eve Mins	Churn
415	?	197.4	n
415	161.6	195.5	n
415	?	121.2	n
408	299.4	61.9	n
415	166.7	148.3	y
510	223.4	220.6	n
510	218.2	348.5	n
415	157	103.1	n
408	184.5	351.6	n
415	258.6	222	n

File **DATA SET**

Column ATTRIBUTE; property or characteristic of an object that may vary, either from one object to another or from one time to another.

EVE MINS can **VARY FROM RECORD TO RECORD**

After two days you receive an **EMAIL MESSAGE FROM YOUR FRIEND** with attached a txt file named churn.



You download and open the **CHURN.TXT FILE AND INSPECT THE FIRST 4 COLUMNS AND THE FIRST 10 LINES:**

Area Code	Day Mins	Eve Mins	Churn
415	?	197.4	n
415	161.6	195.5	n
415	?	121.2	n
408	299.4	61.9	n
415	166.7	148.3	y
510	223.4	220.6	n
510	218.2	348.5	n
415	157	103.1	n
408	184.5	351.6	n
415	258.6	222	n

File **DATA SET**

Column ATTRIBUTE; property or characteristic of an object that may vary, either from one object to another or from one time to another.

AREA CODE takes integer values

EACH ATTRIBUTE IS OF A TYPE, and the TYPE SHOULD TELL US WHAT PROPERTIES OF THE ATTRIBUTE ARE REFLECTED IN THE VALUES USED TO MEASURE IT.

Area Code	Day Mins	Eve Mins	Churn	Int'l Plan	VMail Plan	Day Calls	Night Calls	Night Charge	Intl Calls	State	Phone
415	265.1	197.4	n	0	1	110	91	11.01	3	?	382-4657
415	161.6	195.5	n	0	1	123	103	11.45	3	OH	371-7191
?	243.4	121.2	n	0	0	114	104	7.32	5	NJ	358-1921
408	299.4	61.9	n	?	0	71	89	8.86	7	OH	375-9999
415	166.7	148.3	y	1	0	113	121	8.41	3	OK	330-6626
510	223.4	220.6	n	1	0	98	118	9.18	?	AL	?
510	218.2	348.5	n	0	1	88	118	9.57	7	MA	355-9993
415	157	103.1	n	1	0	?	96	9.53	6	MO	329-9001
408	184.5	351.6	n	0	0	97	90	9.71	4	LA	335-4719
415	?	222	n	1	1	84	97	14.69	5	WV	330-8173

Knowing the **TYPE OF AN ATTRIBUTE** is important because it **TELLS US WHICH PROPERTIES OF THE MEASURED VALUES ARE CONSISTENT WITH THE UNDERLYING PROPERTIES OF THE ATTRIBUTE**, and therefore, it allows us to **AVOID FOOLISH ACTIONS**, such as computing the average value of **Area Code**.

EACH ATTRIBUTE IS OF A TYPE, and the TYPE SHOULD TELL US WHAT PROPERTIES OF THE ATTRIBUTE ARE REFLECTED IN THE VALUES USED TO MEASURE IT.

Area Code	Day Mins	Eve Mins	Churn	Int'l Plan	VMail Plan	Day Calls	Night Calls	Night Charge	Intl Calls	State	Phone
415	265.1	197.4	n	0	1	110	91	11.01	3	?	382-4657
415	161.6	195.5	n	0	1	123	103	11.45	3	OH	371-7191
?	243.4	121.2	n	0	0	114	104	7.32	5	NJ	358-1921
408	299.4	61.9	n	?	0	71	89	8.86	7	OH	375-9999
415	166.7	148.3	y	1	0	113	121	8.41	3	OK	330-6626
510	223.4	220.6	n	1	0	98	118	9.18	?	AL	?
510	218.2	348.5	n	0	1	88	118	9.57	7	MA	355-9993
415	157	103.1	n	1	0	?	96	9.53	6	MO	329-9001
408	184.5	351.6	n	0	0	97	90	9.71	4	LA	335-4719
415	?	222	n	1	1	84	97	14.69	5	WV	330-8173

Attributes as **INTL CALLS** have many of the properties of numbers.

It makes sense to **COMPARE AND ORDER RECORDS BY INTL CALLS**, as well as to talk about the **DIFFERENCES AND RATIOS OF INTL CALLS**.

EACH ATTRIBUTE IS OF A TYPE, and the TYPE SHOULD TELL US WHAT PROPERTIES OF THE ATTRIBUTE ARE REFLECTED IN THE VALUES USED TO MEASURE IT.

Area Code	Day Mins	Eve Mins	Churn	Int'l Plan	VMail Plan	Day Calls	Night Calls	Night Charge	Intl Calls	State	Phone
415	265.1	197.4	n	0	1	110	91	11.01	3	?	382-4657
415	161.6	195.5	n	0	1	123	103	11.45	3	OH	371-7191
?	243.4	121.2	n	0	0	114	104	7.32	5	NJ	358-1921
408	299.4	61.9	n	?	0	71	89	8.86	7	OH	375-9999
415	166.7	148.3	y	1	0	113	121	8.41	3	OK	330-6626
510	223.4	220.6	n	1	0	98	118	9.18	?	AL	?
510	218.2	348.5	n	0	1	88	118	9.57	7	MA	355-9993
415	157	103.1	n	1	0	?	96	9.53	6	MO	329-9001
408	184.5	351.6	n	0	0	97	90	9.71	4	LA	335-4719
415	?	222	n	1	1	84	97	14.69	5	WV	330-8173

The following **PROPERTIES (OPERATIONS) OF NUMBERS ARE TYPICALLY USED TO DESCRIBE ATTRIBUTES**

DISTINCTNESS	= and ≠
ORDER	<, ≤, > and ≥
ADDITION	+ and −
MULTIPLICATION	* and /

The above properties allow to define four **TYPES OF ATTRIBUTES**

CATEGORICAL (QUALITATIVE)

NUMERIC (QUANTITATIVE)

The above properties allow to define four **TYPES OF ATTRIBUTES**

ATTRIBUTE TYPE		DESCRIPTION	EXAMPLES	OPERATIONS
CATEGORICAL (QUALITATIVE)	NOMINAL	The values of a nominal attribute are just different names; i.e., nominal values provide only enough information to distinguish one object from another (=, ≠).	Area Code	mode
			Churn	entropy
			State	contingency
			eye color	
			gender	
NUMERIC (QUANTITATIVE)				

The above properties allow to define four **TYPES OF ATTRIBUTES**

ATTRIBUTE TYPE		DESCRIPTION	EXAMPLES	OPERATIONS
CATEGORICAL (QUALITATIVE)	NOMINAL	The values of a nominal attribute are just different names; i.e., nominal values provide only enough information to distinguish one object from another ($=$, $\neq$).	Area Code	mode
			Churn	entropy
			State	contingency
			eye color	
			gender	
	ORDINAL	The values of an ordinal attribute provide enough information to order objects ($<$, $>$).	{bad, good, excellent}	median
			grades	percentiles
			street numbers	rank correlation
				run tests
			sign tests	
NUMERIC (QUANTITATIVE)				

The above properties allow to define four **TYPES OF ATTRIBUTES**

ATTRIBUTE TYPE		DESCRIPTION	EXAMPLES	OPERATIONS
CATEGORICAL (QUALITATIVE)	NOMINAL	The values of a nominal attribute are just different names; i.e., nominal values provide only enough information to distinguish one object from another ($=$, $\neq$).	Area Code	mode
			Churn	entropy
			State	contingency
			eye color	
			gender	
	ORDINAL	The values of an ordinal attribute provide enough information to order objects ($<$, $>$).	{bad, good, excellent}	median
			grades	percentiles
			street numbers	rank correlation
				run tests
NUMERIC (QUANTITATIVE)	INTERVAL	For interval attributes, the difference between values are meaningful, i.e., a unit of measurements exists ($+$, $-$).	calendar dates	mean
			temperature in Celsius or Fahrenheit	standard deviation
				Pearson's correlation
				t and F tests

The above properties allow to define four **TYPES OF ATTRIBUTES**

ATTRIBUTE TYPE	DESCRIPTION	EXAMPLES	OPERATIONS
CATEGORICAL (QUALITATIVE)	NOMINAL The values of a nominal attribute are just different names; i.e., nominal values provide only enough information to distinguish one object from another ($=$, $\neq$).	Area Code	mode
		Churn	entropy
		State	contingency
		eye color	
		gender	
	ORDINAL The values of an ordinal attribute provide enough information to order objects ($<$, $>$).	{bad, good, excellent}	median
		grades	percentiles
		street numbers	rank correlation
			run tests
			sign tests
NUMERIC (QUANTITATIVE)	INTERVAL For interval attributes, the difference between values are meaningful, i.e., a unit of measurements exists ($+$, $-$).	calendar dates	mean
		temperature in Celsius or Fahrenheit	standard deviation
			Pearson's correlation
			t and F tests
	RATIO For ratio attributes, both differences and ratios are meaningful, ($*$, $/$).	Day Mins	geometric mean
		Eve Mins	harmonic mean
		monetary quantities	percentiles
		length	variation
		electrical current	

An independent way of **DISTINGUISHING** between **ATTRIBUTES** is **BY THE NUMBER OF VALUES THEY CAN TAKE**.

✓ **DISCRETE;** A discrete attribute has a **FINITE OR COUNTABLY INFINITE SET OF VALUES**. It can be

- **CATEGORICAL** (Transaction_ID, ZIP codes, Area Code)
- **NUMERIC** (Day Mins, Eve Mins, counts)
- **BINARY**, Churn special case assuming 2 values (male/female, yes/no)

✓ **CONTINUOUS;** A continuous attribute is one whose **VALUES ARE REAL NUMBERS**.

Examples include attributes such as

TEMPERATURE, HEIGHT, WEIGHT

Typically represented as floating points variables.